Chapter III.9

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II.3.16. *Noetherian Induction.* Let X be a noetherian topological space, and let $\mathscr{P}$ be a property of closed subsets of X. Assume that for any closed subset Y of X, if $\mathscr{P}$ holds for every proper closed subset of Y, then $\mathscr{P}$ holds for Y. (In particular, $\mathscr{P}$ must hold for the empty set.) Then $\mathscr{P}$ holds for X.

Proof. Noetherianity allows us to write $X = \bigcup_{i=1}^{s} X_i$ for X_i irreducible, with some integer $n = \sum_{i=1}^{s} \dim X_i$. We induct on (n, s) with the lexicographic ordering. If $n = s = 0$, then $X = \emptyset$, so $\mathscr{P}$ holds for X. If $n = 0$ and $s > 0$, then $X = \{x_1, \ldots, x_s\}$ is the union of finitely many closed points. Then every proper subset of $X' \subset S$ is a union of $s' < s$ closed points, so that $\mathscr{P}$ holds for X' by induction, whence $\mathscr{P}$ holds for X.

Now, write $X = \bigcup_{i=1}^{s} X_i$, reordering so that $\dim X_i > 0$ for $1 \leqslant i \leqslant r$ and $X_j = \{x_j\}$ are closed points for $r < j \leqslant s$. Take a proper closed subset $X' \subsetneq X$. We have an irreducible decomposition

$$X' = \bigcup_{i=1}^{s} X' \cap X_i$$

for X', with some $X' \cap X_i$ empty. Let s' be the number of nonempty $X' \cap X_i$, and $n' = \sum_{i=1}^{s} \dim X' \cap X_i$.

Take some i such that $X_i \cap X' \subsetneq X_i$ is a proper closed subset. If $i > r$, then $X_i \cap X' = \emptyset$ gives us $s' < s$, so $\mathscr{P}$ holds for X' by induction. Otherwise we have $\dim X_i \cap X' < \dim X_i$, so that $n' < n$, whence $\mathscr{P}$ holds for X' by induction once more. We conclude that, in fact, $\mathscr{P}$ holds for X □

II.3.17 *Zariski Spaces.* A topological space X is a *Zariski space* if it is noetherian and every (nonempty) closed irreducible subset of X has a unique generic point.

(a) Show that if X is a noetherian scheme, then $\mathrm{Sp}(X)$ is a Zariski space.

(b) show that any minimal nonempty closed subset of a Zariski space onsists of one point. We call these *closed points.*

(c) Show that a Zariski space X satisfies the axiom T_0: given any two distinct points of X, there is an open set containing one but not the other.

(d) If X is an irreducible Zariski space, then its generic point is contained in every nonempty open subset of X.

(e) If x_0, x_1 are points of a topological space X ,and if $x_0 \in \{x_1\}^-$, then we say x_1 *specializes* to x_0, written $x_1 \rightsquigarrow x_0$. We also say x_0 is a *specialization* of x_1, or that x_1 is a *generalization of* x_0. Now, let X be a Zariski space. Show that the minimal points, for the partial ordering determined by $x_1 > x_0$ if $x_1 \rightsquigarrow x_0$, are the closed points, and the maximal points are the generic points of the irreducible components of X. Show also that a closed subset contains every specialization of any of its points. (We say closed subsets are *stable under specialization.*) Similarly, open subsets are *stable under generalization.*

(f) Show that the maximal spectrum of a ring A is Zariski if and only if it is equal to $\operatorname{Spec} A$.

Proof. (a) Trivial, given (II, Ex. 2.9).

(b) Let X be a Zariski space, and let $\emptyset \neq Y \subset X$ be a minimal nonempty closed subset of X. In particular, Y is irreducible, and thus contains a unique generic point $\zeta \in Y$. Minimality implies $\{y\}^- = Y$ for all $y \in Y$, but then uniqueness implies $Y = \{\zeta\}$, as desired.

(c) Let $x, y \in X$ be points such that $x \in U$ open implies $y \in U$. Then $y \in C$ closed implies $x \in C$, so that $\{y \in C \text{ closed}\} \subset \{x \in C \text{ closed}\}$. Hence,

$$\{x\}^- = \bigcap_{x \in C} C \subset \bigcap_{y \in C} C = \{y\}^-.$$

If $x, y \in X$ have $x \in U$ if and only if $y \in U$, then $\{x\}^- = \{y\}^-$ implies $x = y$.

(d) If $\emptyset \neq U \subset X$ is a nonempty open set, then $x \notin U$ implies $\{x\}^- \subset U^c \subsetneq X$ is proper, and hence x is not generic.

(e) Let $x \in X$ be minimal. $y \in \{x\}^-$ implies $x \rightsquigarrow y$, so that $y = x$. Hence, x is closed. Now, let $\zeta \in X$ be maximal. $\{\zeta\}^- \subset \{y\}^-$ implies $y \rightsquigarrow \zeta$, so that $y = \zeta$. Each irreducible closed subset of X has a generic point by definition, so we have that $\{\zeta\}^-$ is indeed an irreducible component of X.

Let $x \in C$ be contained in a closed set C, and take y such that $x \rightsquigarrow y$. Then $y \in \{x\}^- \subset C$, as desired. Now, we claim that $S \in X$ is closed under generalization if and only if S^c is closed under specialization. If $x \in S$ and $y \rightsquigarrow x$, then $y \in S^c$ and S^c closed under specialization would imply $x \in S^c$, a contradiction. The converse is identical, and our result for open sets follows.

(f) One direction is clear. □

II.3.18 *Constructible Sets.* Let X be a Zariski topological space. A *constructible subset* is a subset which belongs to the smallest family $\mathfrak{F}$ such that (1) every open subset is in $\mathfrak{F}$, (2) a finite intersection of elements of $\mathfrak{F}$ is in $\mathfrak{F}$, and (3) the complement of any element of $\mathfrak{F}$ is in $\mathfrak{F}$.

(a) A subset of X is *locally closed* if it is the intersection of an open subset with a closed subset. Show that a subset of X is constructible if and only if it can be written a finite disjoint union of locally closed subsets.

(b) Show that a constructible subset of an irreducible zariski space X is dense if and only if it contains the generic point. Furthermore, in that case it contains a nonempty open subset.

(c) A subset S of X is closed if and only if it is constructible and stable under specialization. Similarly, a subset T of X is open if and only if it is constructible and stable under generalization.

(d) If $f : X \to Y$ is a continuous map of Zariski spaces, then the inverse image of any constructible set of Y is constructible in X.

Proof. (a) Write $\mathfrak{L}$ for the family of finite disjoint unions of locally closed subsets. It is clear that $\mathfrak{L} \subset \mathfrak{F}$, so it suffices to show that $\mathfrak{F} \subset \mathfrak{L}$. We will do so by showing $\mathfrak{L}$ satisfies (1-3), and conclude our result via minimality.

(1) is trivial. If $X = X_1 \cap X_2$ for $X_i \in \mathfrak{L}$, then we can write

$$X_i = U_{i1} \cap C_{i1} \sqcup \cdots \sqcup U_{im} \cap C_{im} \quad (i = 1, 2),$$

adding null sets if necessary so that each is a union of m locally closed sets. Then

$$\begin{aligned} X &= \left(\bigsqcup_{j=1}^{m}(U_{1j}\cap C_{1j})\right)\cap\left(\bigsqcup_{j=1}^{m}(U_{2j}\cap C_{2j})\right) \\ &= \bigsqcup_{1\leqslant j,k\leqslant m}(U_{1j}\cap U_{2k})\cap(C_{1j}\cap C_{2k})\in\mathfrak{L}, \end{aligned}$$

Since the union is clearly disjoint. Arbitrary finite intersections follow by induction, giving us (2).

Suppose

$$X = U_1\cap C_1\sqcup\cdots\sqcup U_m\cap C_m\in\mathfrak{L}.$$

Then

$$\begin{aligned} X^c &= U_1\cap C_1\sqcup\cdots\sqcup U_m\cap C_m \\ &= (U_1^c\cup C_1^c)\cap\cdots\cap(U_m^c\cup C_m^c) \\ &= ((U_1\cap C_1^c)\sqcup U_1^c)\cap\cdots\cap((U_m\cap C_m^c)\sqcup U_m^c)\in\mathfrak{L} \end{aligned}$$

by (2), whence we $\mathfrak{L}$ satisfies (3). We conclude that in fact $\mathfrak{F}=\mathfrak{L}$

(b) Let X be an irreducible Zariski space with generic point ζ, and let $Y\in\mathfrak{F}$ be constructible and dense, i.e. $Y^-=X$. By (a), we can write $Y=\bigsqcup_{i=1}^n(U_i\cap C_i)\in\mathfrak{F}$ for $U_i\neq\emptyset$ open and $C_i\neq\emptyset$ closed. Then certainly $X=Y^-\subset\bigcup_{i=1}^n C_i$, so X irreducible implies $X=C_i$ for some i, say $X=C_1$. In particular, $\zeta\in C_1$. Further, we have $\zeta\in U_1$, as otherwise $\zeta\in U_1^c\subsetneq X$ contradicts genericity of ζ. We conclude that, $\zeta\in U_1=C_1\cap U_1\subset Y$, whence both results follow.

(c) (a) and (Ex. 3.17e) give us one direction, while the proof of the latter gives us that S is closed under specialization if and only if S^c is closed under generalization. Now, let $S\in\mathfrak{F}$ be constructible and closed under specialization. It suffices to show that S is closed. Suppose first that $S=U\cap C\neq\emptyset$ for C irreducible with generic point ζ. Then (b) implies $\zeta\in S$, whence $S=C$ by stability under generalization.

Now, take $S=\bigcup_{i=1}^n U\cap C_i$ for C_i irreducible and $U\cap C_i\neq\emptyset$. We claim that each $U\cap C_i$ is closed under specialization. Let $x\in U\cap C_i$ and take $x\rightsquigarrow y$. Then $y\in C_i$ and $y\in U\cap C_j\subset U$ for some j, so, indeed, $y\in U\cap C_i$. It follows that the result holds when $S=U\cap C$ for U open and C closed.

Finally, (a) allows us to write $S=\bigsqcup_{i=1}^n(U_i\cap C_i)$ for U_i open and C_i closed. We claim that each $U_i\cap C_i$ is closed under specialization, whence the result would follow from our previous claim. Let $x\in U_i\cap C_i$ and take $x\rightsquigarrow y$. Then $y\in U_j\cap C_j$ for some j, so closure of open sets under generalization gives us $x\in U_j$, Suppose, by way of contradiction, that $i\neq j$. Disjointness implies $y\in U_i^c$ and $x\in C_j^c$. Then

$$\{y\}^-\subset\{x\}^-\cap C_j\subset\{x\}^-\setminus\{x\}$$

FINISH THIS!!

(d) If $S = \bigsqcup_{i=1}^{n}(U_i \cap C_i)$ is constructible in Y, then

$$\begin{aligned} f^{-1}(S) &= \bigcup_{i=1}^{n} f^{-1}(U_i \cap C_i) \\ &= \left(\bigcap_{i=1}^{n} (f^{-1}(U_i) \cap f^{-1}(C_i))^c \right)^c \end{aligned}$$

is constructible in X. □

II.3.19 Let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Then the image of any constructible subset of X is a constructible subset of Y. Prove this theorem in the following steps.

(a) Reduce to showing that $f(X)$ itself is constructible, in the case where X and Y are affine, integral noetherian schemes, and f is a dominant morphism.

(b) In that case, show that $f(X)$ contains a nonempty open subset of Y by using the following result from commutative algebra: let $A \subset B$ be an inclusion of noetherian integral domains, with B a finitely generated A-algebra. Then, given a nonzero element $b \in B$, there is a nonzero element $a \in A$ with the following property: if $\phi : A \to K$ is any homomorphism of A to any algebraically closed field K, such that $\phi(a) \neq 0$, then ϕ extends to a homomorphism ϕ' of B into K, such that $\phi'(b) \neq 0$. [*Hint*: Prove the algebraic result by induction on the number of generators of B over A, with a direct proof for B cyclic over A. In the application, take $b = 1$.]

(c) Now use noetherian induction on Y to complete the proof.

(d) Give some examples of morphisms $f : X \to Y$ of varieties over an algebraically closed field k, to show that $f(X)$ need not be either open or closed.

Proof. (a) □

9.1. A flat morphism $f : X \to Y$ of finite type of noetherian schemes is open, i.e. for every open subset $U \subset X$, $f(U)$ is open in Y. [*Hint*: show that $f(U)$ is constructible and stable under generalization (II, Ex. 3.18) and (II, Ex. 3.19).]

Proof. $f(U)$ is constructible by (II, Ex. 3.19), so it suffices to show it is stable under generalization by (II, Ex. 3.18c). Suppose $y' \rightsquigarrow y = f(x)$ for some $x \in X$, so y' is the image of some prime $\mathfrak{q}' \in \operatorname{Spec} \mathcal{O}_y$ under the inclusion $\operatorname{Spec} \mathcal{O}_y \hookrightarrow Y$. The map $f^\sharp : \mathcal{O}_y \to \mathcal{O}_x$ is flat, so (AM 3.18) gives us that it is in fact surjective.

$$\begin{array}{ccc} X & \xrightarrow{\quad f \quad} & Y \\ \uparrow & & \uparrow \\ \operatorname{Spec} \mathcal{O}_x & \xrightarrow{\quad f \quad} & \operatorname{Spec} \mathcal{O}_y \end{array}$$

In particular, it supplies us with a prime $\mathfrak{p}' \in \operatorname{Spec} \mathcal{O}_x$ such that $f^\sharp(\mathfrak{p}') = \mathfrak{q}'$. The image $x' \in X$ of $\mathfrak{p}'$ gives us $y' = f(x') \in f(X)$, as desired. □

9.3. Some examples of flatness and nonflatness.

(a) If $f : X \to Y$ is a finite surjective morphism of nonsingular varieties over an algebraically closed field k, then k is flat.

(b) Let X be a union of two planes meeting at a point, each of which maps isomorphically to a plane Y. Show that f is not flat. For example, let $Y = \operatorname{Spec} k[x, y]$ and $X = \operatorname{Spec}[x, y, z, w]/(z, w) \cap (x + z, y + w)$.

(c) Again let $Y = \operatorname{Spec} k[x, y]$, but take $X = \operatorname{Spec} k[x, y, z, w]/(z^2, zw, w^2, xz - yw)$. Show that $X_{\text{red}} \cong Y$ has no embedded points, but that f is not flat.

9.4. *Open Nature of Flatness.* Let $f : X \to Y$ be a morphism of finite type of noetherian schemes. Then $\{x \in X | f \text{ if flat at } x\}$ is an open subset of X (possibly empty).

Proof. □

9.5.

9.7. Let $Y \subset X$ be a closed subscheme, where X is a scheme of finite type over a field k. Let $D = k[t]/t^2$ be the ring of dual numbers, and define an *infinitesimal deformation of Y as a closed subscheme of X*, to be a closed subscheme $Y' \subset X \times_k D$, which is flat over D and whose closed fibre is Y. Show that they are classified by $H^0(Y, \mathscr{N}_{Y/X})$, where

$$\mathscr{N}_{Y/X} = \mathcal{H}om_{\mathcal{O}_X}(\mathscr{I}_Y/\mathscr{I}_Y^2, \mathcal{O}_Y).$$

Proof. We consider first the affine case, so that $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} A/I$. Then $X \times_k D = \operatorname{Spec} B$ with $A[t]/t^2$, and $Y' = \operatorname{Spec} B/I'$ for

$$I' = (a_1 + b_1 t, \ldots, a_r + b_r t, b_{r+1} t, \ldots, b_s t)$$

with $b_1, \ldots, b_s \in A$ and $(a_1, \ldots, a_r) = I$. Hence, the coordinate ring of the closed fibre is given by

$$\frac{B}{I'} \otimes_{k[t]/t^2} \frac{k[t]/t^2}{(t)} \cong \frac{B/(t)}{I'/(t)} \cong \frac{A}{I}.$$

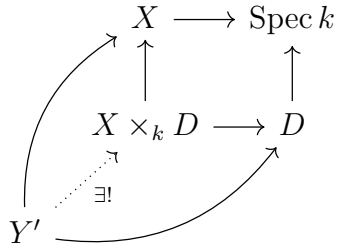
Note that any I' can be written in this way without duplicate a_i, as $a_i + b_i' t = a_i + b_i t + (b_i' - b_i)t$. It suffices to find a condition for $Y' \to D$ to be flat.

Recall that a morphism of affine schemes is flat if and only if the associate map of rings is flat. (9.1Aa) gives us that $k[t]/t^2 \to B/I'$ is flat if and only if the multiplication map $\mathfrak{a} \otimes B/I' \to B/I'$ is injective for every finitely generated ideal $\mathfrak{a} \trianglelefteq k[t]/t^2$. There are only two ideals of $k[t]/t^2$, and $(1) \otimes B/I' \to B/I'$ is an isomorphism. Thus, we need only consider $(t) \otimes B/I' \to B/I'$.

First, we note that $t \otimes bt = 0$. Then if $b \in B/I' \setminus (t)/I'$

Now, consider $\mathscr{N}_{Y/X} = \operatorname{Hom}_A(I/I^2, A/I)$

The universal property of $X \times_k D$ gives us that maps $Y' \to X \times_k D$ are in correspondence with commuting diagrams of the following form.



□

9.10. A scheme X_0 over a field k is *rigid* if it has no infinitesimal deformations.

(a) Show that $\mathbb{P}^1_k$ is rigid, using (9.13.2).

Proof. It suffices via (9.13.2) to show that $H^1(X, \mathscr{T}_X) = 0$, with $X = \mathbb{P}^1_k$.

$$\begin{aligned} \mathscr{T}_X &= \Omega_X^\vee \\ &= \omega_X^\vee \\ &\cong \mathcal{O}(2). \end{aligned}$$

Then (7.1) and (II, 5.1) give us

$$\begin{aligned} H^1(X, \mathscr{T}_X)^\vee &\cong \operatorname{Hom}(\mathscr{T}_X, \omega_X) \\ &\cong \Gamma(X, \mathcal{O}(-4)) \\ &= 0, \end{aligned}$$

since $-4 < 0$. □